DUBLIN BUSINESS SCHOOL

MSc Artificial Intelligence

Natural Language Processing | B9AI006

University Knowledge Base Chatbot

A Retrieval-Augmented Generation System

CA02 Individual Report — Task 2

Student Name:	Bhargava Koya
Student ID:	20075511
Module:	Natural Language Processing (B9AI006)
Lecturer:	Terri Hoare
Assessment:	CA2 — Individual Report (Task 2)
Programme:	MSc Artificial Intelligence
Institution:	Dublin Business School
Submission Date:	19/04/2026

April 2026

1. Introduction

This report constitutes the individual component (Task 2) of the CA02 assessment for the NLP module (B9AI006). The group project involved designing and building a Retrieval-Augmented Generation (RAG) chatbot for a university knowledge base, a system that answers student queries by grounding responses in institutional documents rather than relying on a general-purpose language model's pre-training data.

RAG systems address a fundamental limitation of standalone LLMs: they cannot reliably answer questions about institution-specific policies, deadlines and requirements that were not present in their pre-training data. For a university context, this distinction is critical — a student asking about submission deadlines or late-penalty policies requires precise, document-verified answers, not probabilistic confabulation.

The system architecture operates as follows: PDF and DOCX institutional documents are loaded using LangChain document loaders and segmented into overlapping chunks using RecursiveCharacterTextSplitter (chunk_size=800, overlap=80). Each chunk is embedded using the locally hosted HuggingFace sentence-transformers/all-MiniLM-L6-v2 model (384-dimensional vectors) and stored in a FAISS vector index. At query time, the bi-encoder retrieves fetch_k=15 candidate chunks, a cross-encoder reranker (ms-marco-MiniLM-L-6-v2) then jointly scores each (query, chunk) pair to rerank and select the top_k=5 most relevant chunks, which are injected into a structured prompt for Anthropic Claude 3 Haiku (temperature=0.0).

This report addresses three analytical tasks: ethical considerations (Task 2a), a critique of modelling techniques and libraries (Task 2b), and a performance evaluation of the system (Task 2c).

2. Task 2a — AI Principles: Ethics, Bias, Fairness, Explainability and Interpretability

Criterion 002

2.1 Ethics

The EU AI Act (2024) classifies AI systems operating in educational contexts as high-risk, given their potential to directly affect students' academic outcomes. This chatbot falls squarely within that classification: a student receiving an incorrect answer about a submission deadline or a late-penalty calculation and acting on that information faces real academic harm not merely inconvenience. The ethical obligations arising from this classification are substantial.

Two core ethical principles from the UNESCO Recommendation on the Ethics of AI (2021) — beneficence (acting in users' interests) and non-maleficence (avoiding harm) are operationalised through architectural choices rather than post-hoc policy. The system prompt explicitly instructs the LLM to answer using only the information in the provided context and to state when the answer cannot be found rather than speculating. This constrained generation instruction prioritises accuracy over fluency, reducing the risk that a student acts on a hallucinated policy detail.

Transparency is a second ethical obligation. Article 52 of the EU AI Act requires that AI systems interacting with humans disclose their automated nature. The chatbot interface must make clear to users that they are interacting with an automated system and should verify critical decisions against authoritative human advisors or official institutional sources. Institutional responsibility — the obligation of the university rather than solely the technology provider to ensure response accuracy requires that source documents are version-controlled, reviewed and updated before ingestion into the knowledge base.

2.2 Bias

2.2.1 Document Bias

The knowledge base in this project comprises documents written in formal academic institutional English. Students from non-English-speaking backgrounds, or those less familiar with Irish academic conventions, may phrase queries in registers that diverge from the document's vocabulary, reducing retrieval recall in a manner that correlates with linguistic background. This constitutes a form of indirect discrimination as defined in Barocas and Hardt's Fairness and Machine Learning (2023): a facially neutral system that produces disparate outcomes across demographic groups. Mitigation strategies include diversifying the corpus with FAQ-format documents, plain-language policy summaries, and multilingual query expansion.

2.2.2 Embedding Bias

The sentence-transformers/all-MiniLM-L6-v2 model was pre-trained on general-purpose corpora. Domain-specific institutional terms such as "MIMLO", "QAH", "Ouriginal", or "Brightspace" may be embedded with lower semantic fidelity than everyday vocabulary, creating a systematic retrieval disadvantage for queries that use precise institutional terminology. A student asking about the "QAH Section 5.2.2.6 late penalty" may receive lower-quality retrieval than one asking "what happens if I submit late?" the opposite of the intended behaviour. Fine-tuning the embedding model on university domain text would address this gap.

2.3 Fairness

Algorithmic fairness in this system means that answer quality should be independent of how a question is phrased. A student asking "what happens if I hand in late?" should receive the same factual content as one asking "what is the late submission penalty under QAH Section 5.2.2.6?". This is a group fairness requirement equalised retrieval recall across query-style populations.

In this project, the evaluation set of 51 questions deliberately includes paraphrased variants of key queries to assess whether retrieval precision degrades across phrasing styles. The cross-encoder reranker partially addresses this by jointly reasoning over query and chunk text, reducing dependence on surface-level lexical similarity. A full fairness audit would require systematic paraphrase testing across demographic proxies, a direction for future work.

2.4 Explainability

RAG systems offer a structural explainability advantage over pure generative LLMs: every answer is linked to the specific source chunks that informed it, including document name and page number. This satisfies the right-to-explanation principle underpinning GDPR Article 22 and the EU AI Act's transparency requirements for high-risk systems. Users can interrogate the reasoning pathway inspecting which document passage produced a given answer rather than accepting an answer on faith.

In this system, every answer returned by answer_with_rag() includes a sources list containing the retrieved Document objects with source filename and page number metadata, enabling full answer auditability. This is a meaningful distinction from a standalone LLM, where the answer arrives with no traceable reasoning path.

2.5 Interpretability

The pipeline components are individually interpretable at a component level: the retriever can be audited by examining returned chunks, the prompt template is human-readable and auditable. The LLM's generation is constrained to the retrieved context, making deviations (hallucinations) detectable through source comparison. The two-stage retrieval design bi-encoder for broad recall, cross-encoder for precision reranking allows each stage to be evaluated and debugged independently.

However, the internal attention mechanisms of the transformer architecture underlying both the embedding models and the generative LLM remain opaque. This is the fundamental interpretability ceiling of current neural language models and it applies regardless of whether retrieval augmentation is used. Techniques such as attention visualisation and probing classifiers provide partial interpretability but do not yield fully human-readable explanations of model behaviour.

3. Task 2b — Critique of Modelling Techniques and Libraries Used

Criterion 003

3.1 LangChain

LangChain was selected as the primary orchestration framework because it provides pre-built, composable abstractions for every stage of the RAG pipeline document loading (PyPDFLoader, Docx2txtLoader), text splitting, embedding, vector store interaction and chain construction. This substantially reduces boilerplate code and enables rapid prototyping.

Advantages: LangChain's modular architecture allows each component to be swapped independently replacing the embedding model or vector store requires a single line change. LangChain Expression Language (LCEL), using the | pipe operator, produces pipeline definitions that are readable and auditable, which is valuable both during development and when presenting the system to a non-technical audience. The framework has strong community support and documentation.

Limitations: LangChain's aggressive deprecation cycle (versions 0.0.x to 0.1 to 0.2+ within a single year) introduced breaking API changes mid-project, requiring unplanned refactoring. The requirements.txt pins langchain>=0.3.0,<0.4.0 specifically to avoid compatibility conflicts. Abstraction layers complicate debugging: when retrieval returns poor results, diagnosing whether the fault lies in chunking, embedding, or the FAISS index requires bypassing LangChain's abstractions and inspecting components directly. For production deployments, LlamaIndex which has a more stable API and stronger focus on retrieval strategies would be a considered alternative.

3.2 FAISS (Facebook AI Similarity Search)

FAISS stores document embeddings as dense vectors and retrieves the k nearest neighbours to a query vector using approximate nearest-neighbour (ANN) search. It was used as the first-stage retriever, returning fetch_k=15 candidate chunks for reranking.

Advantages: FAISS provides extremely fast similarity search, even on CPU and supports large-scale indices (millions of vectors). It is fully local with no external API or network latency which is important for data privacy in institutional deployments where sending student queries to third-party services may raise compliance concerns.

Limitations: FAISS is an in-memory store; for very large corpora, RAM becomes a bottleneck. It does not support dynamic document addition, the index must be fully rebuilt when new documents are ingested. It does not natively support hybrid retrieval combining sparse (BM25 keyword) and dense (semantic) search. Hybrid retrieval has been demonstrated to outperform dense-only retrieval on domain-specific corpora where both exact keyword matching and semantic understanding are required (Luan et al., 2021). A production extension of this system would benefit from migrating to Weaviate or Chroma, both of which support hybrid search and dynamic updates.

3.3 HuggingFace Embeddings (sentence-transformers/all-MiniLM-L6-v2)

This model converts text chunks and user queries into 384-dimensional dense vectors, enabling semantic similarity comparisons using cosine distance. At approximately 22MB, it runs efficiently on CPU without requiring any API key or incurring token costs, a deliberate design choice for privacy-sensitive institutional deployments.

Advantages: The model is free, locally executable and provides strong performance for general semantic similarity tasks, with sentence-level representations that capture meaning beyond keyword overlap, as demonstrated by Reimers and Gurevych (2019).

Limitations: The model was not fine-tuned on university-specific institutional language. Domain-specific terms such as "MIMLO", "QAH", or "Brightspace" may be embedded with lower precision than general English vocabulary, creating a semantic gap between query and document chunks for the most specialised institutional queries. A higher-quality embedding alternative OpenAI text-embedding-ada-002 would improve embedding quality but at the cost of per-token API pricing and external data transmission.

3.4 Cross-Encoder Reranker (ms-marco-MiniLM-L-6-v2)

The two-stage retrieval architecture directly addresses a fundamental limitation of bi-encoder systems. In Stage 1, the FAISS bi-encoder retrieves fetch_k=15 candidate chunks by independently embedding both query and document and matching on cosine similarity. This is fast but context-blind: the phrase "submission deadline" has similar embedding representations across all subject PDFs (NLP, ML, Recommender Systems), causing the bi-encoder to retrieve chunks from the wrong module's document.

In Stage 2, the cross-encoder reads each (query, chunk) pair jointly attending over both texts simultaneously. This allows it to recognise that "NLP CA2" in the query is a strong contextual signal for the NLP document, even when the generic phrase "submission deadline" appears in multiple documents. The cross-encoder reranks the 15 candidates and selects the top 5 for the LLM context window.

A concrete demonstration of reranker impact: before enabling the cross-encoder, the query "What is the submission deadline for the NLP CA2 assignment?" consistently retrieved chunks from the ML and Recommender Systems PDFs. After enabling the cross-encoder, retrieval correctly prioritised NLP-specific chunks. This demonstrates that two-stage retrieval is not merely an incremental improvement but is qualitatively necessary for multi-document corpora where the same generic phrases appear across multiple domain-specific documents.

3.5 RecursiveCharacterTextSplitter

This splitter divides documents hierarchically: first on double newlines, then single newlines, then sentences, then words. This preserves semantic units better than fixed-size character splits that may cut sentences or table rows mid-content.

A chunk_size=800 characters was selected to preserve table rows and bullet list items intact critical for assessment policy documents where tabular data (mark allocations, deadline tables) must not be fragmented. The chunk_overlap=80 parameter prevents information loss at chunk boundaries, a standard heuristic in RAG literature (Gao et al., 2023). The optimal chunk size is corpus-dependent: smaller chunks (200–300 chars) improve retrieval precision but risk fragmenting sentences; larger chunks (1000+ chars) provide richer context but introduce retrieval noise. Empirical tuning across chunk sizes using a held-out evaluation set would be the principled approach.

3.6 Anthropic Claude 3 Haiku (Generation LLM)

Claude 3 Haiku was selected as the generative LLM for its balance of speed, cost, and instruction-following quality. At temperature=0.0, the model produces fully deterministic outputs critical for a system where reproducibility and consistency are expected by users querying the same question across multiple sessions.

The system prompt explicitly grounds the model: it is instructed to answer using only the provided context and to respond with a standard refusal message if the answer is not present, rather than hallucinating plausible-sounding but incorrect details. This grounding mechanism is the primary defence against confabulation at the generation stage, as documented by Maynez et al. (2020) in abstractive summarisation systems. The max_tokens=1024 limit keeps answers concise and prevents runaway generation.

3.7 Design Trade-offs

Table 1 below summarises the key design decisions, alternatives considered and the trade-offs involved.

Table 1: RAG System Design Trade-offs

Design Decision	Choice Made	Alternative	Trade-off
Retrieval Type	Dense semantic (FAISS)	Sparse BM25 keyword	Dense handles paraphrase well, sparse handles exact terms and rare words
Embedding Model	all-MiniLM-L6-v2 (local)	OpenAI text-embedding-ada-002	Local: zero cost, full privacy; OpenAI: higher quality but external API dependency

Design Decision	Choice Made	Alternative	Trade-off
Reranking	Cross-encoder reranker	Bi-encoder only	Reranking adds ~23ms but critically improves subject-specific precision in multi-doc corpora
LLM Temperature	0.0 (deterministic)	>0.0 (stochastic)	Temperature 0.0 maximises reproducibility, higher temperature may produce more fluent answers
Top-K Chunks	top_k=5 (post-reranking)	3 or 10	More chunks increase context coverage but introduce noise, fewer are precise but may miss information
Chunk Size	800 characters	300–500 characters	Larger preserves table and list structure; smaller enables more granular retrieval

4. Task 2c — Performance Evaluation of Results

Criterion 004

4.1 Evaluation Methodology

Evaluating RAG systems is an active research challenge, as the quality of generated answers depends on both retrieval quality and generation quality dimensions that are partially independent and require different evaluation approaches. This project implements a two-tier evaluation framework.

4.1.1 Tier 1: Lightweight Proxy Metrics

Proxy metrics require no human-annotated gold answers and can be computed automatically for any test question. Four metrics were used. The Keyword Coverage Score measures the fraction of expected answer keywords present in the generated answer, defined as:

```
keyword_coverage(answer, K) = |{k ∈ K : k ∈ answer}| / |K|
```

For domain-specific factual questions deadlines, policy penalties, mark allocations the correct answer will contain specific, non-paraphraseable terms. A high keyword coverage score indicates the system retrieved and surfaced the relevant information. The Source Citation Rate measures whether at least one grounded source document was returned with each answer. Response Time measures end-to-end pipeline latency in seconds. Answer Word Count measures conciseness.

Limitation: keyword matching cannot distinguish paraphrased correct answers from wrong answers that coincidentally contain expected keywords. Embedding-based semantic similarity scoring (e.g., BERTScore) would be more robust for this purpose.

4.1.2 Tier 2: RAGAS Automated Evaluation

RAGAS (Es et al., 2023) uses an LLM-as-judge approach to evaluate RAG systems without requiring manually annotated gold answers for all metrics. Table 2 below summarises the four RAGAS metrics used.

Table 2: RAGAS Evaluation Metrics

Metric	Definition	Ground Truth Required?
Faithfulness	Fraction of answer claims entailed by retrieved context. Score of 1.0 = fully grounded; 0.0 = hallucination.	No
Answer Relevancy	Semantic alignment between generated answer and original question. Near 1.0 = directly addresses the question.	No
Context Precision	Whether most relevant chunks are ranked highest by the retriever.	Yes
Context Recall	Whether retrieved context covers all information needed to answer the question.	Yes

4.2 Evaluation Results

The evaluation was conducted on a test set of 51 questions spanning five categories: NLP Submission Policy, Assessment Details, Academic Integrity, Late Submission and Technical Requirements. Table 3 presents the proxy metric results alongside a comparison to a no-retrieval baseline (the same LLM answering without retrieved context).

Table 3: Evaluation Results — RAG System vs. No-Retrieval Baseline

Metric	RAG System	Baseline (No Retrieval)
Average Keyword Coverage	60.3%	~34% (estimated)
Source Citation Rate	100%	0%
Average Response Time	1.35 seconds	~0.9 seconds
Total Questions Evaluated	51	51

The 100% source citation rate is a critical result for institutional deployment: every RAG response returns at least one grounded source document, providing full answer auditability a requirement that a standalone LLM cannot satisfy by design. The average keyword coverage of 60.3% indicates that the system correctly surfaces the majority of expected answer terms, though there is meaningful room for improvement.

The response time of 1.35 seconds is acceptable for an asynchronous knowledge base chatbot. The latency is dominated by the Anthropic Claude 3 Haiku API round-trip; the FAISS bi-encoder search typically completes in under 5ms, and the cross-encoder reranking adds approximately 23ms per 15-candidate batch both negligible relative to the LLM API call.

4.3 Key Finding — Reranker Impact on Subject-Specific Retrieval

The most significant qualitative finding concerns the impact of the cross-encoder reranker on subject-specific retrieval precision in a multi-document corpus. Before enabling the cross-encoder, the query "What is the submission deadline for the NLP CA2 assignment?"

consistently retrieved chunks from the ML module PDF and the Recommender Systems PDF rather than the NLP PDF. This occurred because the bi-encoder embeds the query independently of the documents, matching on generic semantic similarity and the phrase "submission deadline" has near-identical embedding representations across all subject PDFs.

After enabling the cross-encoder, the pipeline correctly prioritised NLP-specific chunks. The cross-encoder's joint attention over the full (query, chunk) pair allowed it to recognise that "NLP CA2" in the query is a strong contextual signal for the NLP document context, even when the generic phrase "submission deadline" appears identically across multiple documents. This qualitative distinction demonstrates that two-stage retrieval is not merely an incremental improvement but is architecturally necessary for multi-document institutional corpora.

4.4 Limitations and Recommendations

The evaluation has four material limitations. First, 51 questions is insufficient for statistically significant conclusions; a minimum of 100+ annotated pairs per question category would be required for a robust evaluation. Second, keyword matching is a coarse proxy: it penalises semantically correct paraphrased answers and rewards incorrect answers that happen to contain expected terms. BERTScore or a human evaluation panel would provide a more reliable signal. Third, the single-document corpus does not stress-test cross-document retrieval at scale. Fourth, no adversarial queries, ambiguous compound questions, negation, hypothetical scenarios were included.

Recommendations for future work: implement hybrid search combining dense semantic and sparse BM25 retrieval to improve precision on keyword-heavy queries (exact deadline dates, specific mark allocations); add ground_truth reference answers to enable RAGAS Context Precision and Context Recall metrics; fine-tune the embedding model on university domain text to reduce the institutional terminology gap; and expand the knowledge base to multiple university documents (module guides, student handbook, academic regulations) to create a comprehensive institutional knowledge base.

5. Conclusion

This report has analysed the design, implementation and evaluation of a Retrieval-Augmented Generation chatbot for a university knowledge base through three analytical lenses: AI ethics and principles, modelling technique critique and performance evaluation.

From an ethics and AI principles perspective, the system demonstrates that transparency and explainability can be built into the architecture through source attribution and constrained generation rather than added as post-hoc features. However, the EU AI Act's high-risk classification for educational AI systems imposes ongoing obligations around auditability, disclosure and institutional accountability that extend well beyond the technical implementation. Document bias and embedding bias remain important considerations for any production institutional deployment.

From a modelling perspective, the two-stage retrieval architecture FAISS bi-encoder for broad recall, cross-encoder reranker for subject-specific precision proved qualitatively important for multi-document corpora. The cross-encoder's ability to jointly reason over (query, chunk) pairs resolved retrieval ambiguity that the bi-encoder alone could not address. The choice of a local embedding model (all-MiniLM-L6-v2) over a proprietary API represents a deliberate privacy-utility trade-off appropriate for institutional deployments.

From a performance evaluation perspective, the results 60.3% average keyword coverage, 100% source citation rate, 1.35s average response time provide a meaningful baseline for future improvement. The qualitative finding on reranker impact on subject-specific retrieval is the most significant result and has direct implications for any multi-document institutional knowledge base deployment.

6. References

Anthropic. (2024). Claude 3 model card. Anthropic. https://www.anthropic.com/claude

Barocas, S., & Hardt, M. (2023). Fairness and machine learning: Limitations and opportunities. MIT Press.

Es, S., James, J., Espinosa-Anke, L., & Schockaert, S. (2023). RAGAS: Automated evaluation of retrieval augmented generation. arXiv preprint arXiv:2309.15217.

European Parliament. (2024). Artificial Intelligence Act. Official Journal of the European Union.

Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., & Wang, H. (2023). Retrieval-augmented generation for large language models: A survey. arXiv:2312.10997.

Johnson, J., Douze, M., & Jegou, H. (2019). Billion-scale similarity search with GPUs. IEEE Transactions on Big Data, 7(3), 535–547.

LangChain. (2024). LangChain documentation. https://python.langchain.com

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W., Rocktaschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. Advances in Neural Information Processing Systems, 33, 9459–9474.

Luan, Y., Eisenstein, J., Toutanova, K., & Collins, M. (2021). Sparse, dense, and attentional representations for text retrieval. Transactions of the Association for Computational Linguistics, 9, 329–345.

Maynez, J., Narayan, S., Bohnet, B., & McDonald, R. (2020). On faithfulness and factuality in abstractive summarization. Proceedings of ACL 2020.

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using siamese BERT-networks. Proceedings of EMNLP 2019.

Trivedi, H., Balasubramanian, N., Khot, T., & Sabharwal, A. (2022). Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. Proceedings of ACL 2023.

UNESCO. (2021). Recommendation on the ethics of artificial intelligence. United Nations Educational, Scientific and Cultural Organization.